

Validating SDMs

Bob O'Hara

Abstract

Introduction

- SDMs often used. Validation, i.e. how good are they, is an issue. (A. Lee-Yaw et al. (2022)).
- Current validation often on same data, which can lead to over-fitting. Also, metrics often wrong: because it is the same data, will replicate biases. Thus TSS should be FSS in this context.
- A better approach is to validate on external data. Could calculate AUC, TSS etc. but the validation data will be imperfect too. Here we take a slightly different approach that can be adapted to different data types.

Smith and Levine (2025) showed that Yackulic et al. (2013) were right: you shouldn't think at all species are equal. But...

Standards paper shows almost no improvement, not clear there has been any since.

The same data was used by Valavi et al. (2022) to compare the performance of different models. Here our focus is on the comparisons that can and should be made, and what the results tell us.

Theory

Most data is modelled as presence-only, so we will follow that here but indicate the adjustments needed if the data are of a different type. For presence-only data the underlying theory can be derived from point processes (e.g. Fithian and Hastie (2013), Renner and Warton (2013), Warton and Shepherd (2010), Aarts, Fieberg, and Matthiopoulos (2012)). The actual distribution is viewed as an intensity surface, with a higher intensity where a species is more likely to be present. The data is made up of locations of observations of the species of interest, along with locations in the same region where the species was not observed. These latter points are usually called "pseudo-absences" in the SDM world. In a point process approach they are

seen as integration points - to calculate the likelihood for the presences we should integrate the intensity over the full space.

Mathematically, we can say that the intensity surface is $\lambda(\mathbf{s})$, where $\mathbf{s}$ is the location. We assumed that it is affected by the environment in a way that can be written additively:

$$\log \lambda(\mathbf{s}) = \alpha + \sum_i \beta_i X_i(\mathbf{s}) = \alpha + \eta(\mathbf{s})$$

The idea is that even if the effect of a covariate is not additive, we can transform the covariate to be several **features**, $X_i(\mathbf{s})$. MaxEnt and GAMs, for example, use different features to create flexible non-linear transformations.

Of concern here is the model for the intensity surface. It is assumed that it is affected by several environmental covariates in an additive way. Specifically, of $\lambda(\mathbf{s})$ is the intensity at point $\mathbf{s}$, the assumption is that $\log \lambda(\mathbf{s}) = \alpha + \sum_i \beta_i X_i(\mathbf{s}) = \alpha + \eta(\mathbf{s})$ where $X_i(\mathbf{s})$ is a *feature*, i.e. a function of the environmental covariates. For simplicity we can think of this as an environmental covariate, but it could also be a non-linear term, such as a quadratic or the product of two environmental covariates (i.e. an interaction). This same approach is used in GLMs, MaxEnt, GAMs and many other methods. The β_i s are standard regression coefficients, and α is an intercept. This is not normally of interest for presence-only data, as it is determined by the amount of data, i.e. the sampling effort. Thus the focus is on $\eta(\mathbf{s})$.

If the model is correct, then the number of individuals of a species in an area A would follow a Poisson distribution with mean $A\lambda(\mathbf{s})$, where A is a constant that will depend on sampling effort. The probability of an absence is then the probability of zero individuals, i.e. $e^{-A\lambda(\mathbf{s})}$. This leads to a model for presence/absence where $P(Z = 1) = 1 - e^{-\alpha_c - \eta(\mathbf{s})}$, which is equivalent to a GLM with a binomial response and a cloglog link function ($\log(-\log(1 - P(Z = 1))) = \alpha_c + \eta(\mathbf{s})$). An alternative is to use a logistic regression (Elith et al. (2011)), i.e. $\log P(Z = 1)/(1 - P(Z = 1)) = \alpha_l + \eta(\mathbf{s})$, with a derivation based on averaging over possible distributions of the covariates and response (Phillips and Dudík (2008)).

The purpose of laying this out is to suggest an approach to model validation. If we fit a model to presence-only data, we get estimates for $\eta(\mathbf{s}) = \sum_i \beta_i X_i(\mathbf{s})$. We can then use this to calculate predicted probabilities (up to an intercept) for a new presence/absence data set. We can then use these predictions in a GLM with a cloglog or logit link. If the model is correct, the slope should equal 1.

Although we are not primarily interested in the intercept, except to note that it should not equal 0 (as pointed out by both Yackulic et al. (2013) and Smith and Levine (2025)), we might expect that if we look at the estimates for the fitting and validation models across species from the same data sets, they should be correlated. i.e. a species that is more common in the presence only data set may also be more common in the presence-absence data, because it may be more common or be easier to spot.

Effects of Uncertainty

Assume we have a binary response, Y_i , with $g(P(Y_i = 1)) = \eta_i = \alpha_d + \gamma_i$. $g()$ is a link function, such as a logit or cloglog link. We will assume $E(\gamma_i) = 0$, so that α_d is the mean prevalence in the data set d . The environmental effects are modelled through γ_i (typically as $\gamma_i = \sum_j \beta_j x_{ij}$ where x_{ij} is feature j for location i).

A useful way of conceptualising this model is as a threshold model. We assume that we have a latent variable, λ_i . If $\lambda_i < 0$ we have $Y_i = 1$. So a higher mean for λ_i would mean a greater probability of presence.

We can link this to our model by writing $\lambda_i = \alpha_d + \gamma_i + \epsilon_i$, so α is the overall mean, γ_i is the systematic deviation from that mean (i.e. the relative suitability), and ϵ_i is the random component, due to stochasticity, or unmeasured effects such as biotic interactions. Now, saying that $Y_i = 1$ if $\lambda_i < 0$ is the same as saying that $Y_i = 1$ if $\epsilon_i < \eta_i$. Mathematically, $P(\epsilon_i < \eta_i)$ is the cumulative density function of ϵ_i calculated at η_i , so that $g()$ - the link function - is just the cumulative density function assuming some distribution.

The “usual” link functions for binary data are the logit, probit and cloglog, which correspond to the logistic, normal and Gumbel distribution respectively (REF). It is important here to fix the variance of ϵ_i . If instead it has a standard deviation σ , then $\epsilon_i < \eta_i$ is the same as $\epsilon_i/\sigma < \eta_i/\sigma = \sum_j \beta_j x_{ij}/\sigma$. In other words, the effect of changing the variance of ϵ is the same as changing the strengths of the environmental effects.

If the point process model underlying MaxEnt is correct, then the model for presence absence data is a generalised linear model with a cloglog link function (REF). However, a maximum entropy approach suggests a logit link (REF) as an alternative. Whichever link function is chosen, the model is the same as above, but with a different intercept, α_d . The intercept is affected by both the mean prevalence of a species and the average detectability.

What happens when the model is not correct? We can get some intuition by adding an extra sampling variation term, δ_i :

$$\lambda_i = \alpha_d + \gamma_i + \delta_i + \epsilon_i$$

We will assume it is random, with some variance τ^2 , and uncorrelated with γ_i . If we first look at this variation in the PO data for fitting, we see that the model can be written as $\lambda_i = \alpha_d + \gamma_i + (\delta_i + \epsilon_i)$, where the term in brackets has a variance of $\tau^2 + 1$. As this is (by definition) not correlated with γ_i , we can look at it as random, so it inflates the variance, and thus (as argued above), reduces the estimated regression coefficients by $\tau^2 + 1$.

If we then use these same coefficients to explain the PA data, then the prediction will also be reduced by $\tau^2 + 1$. Thus our regression coefficient will be above 1. Conversely, if our model for the PO data is correct, but our PA data has this extra variation, the regression coefficient will

be below 1. These effects are demonstrated in Figure 1 for the Swiss species 04. As the extra variation increases, i.e. the PO data becomes worse, the slope tends to increase.

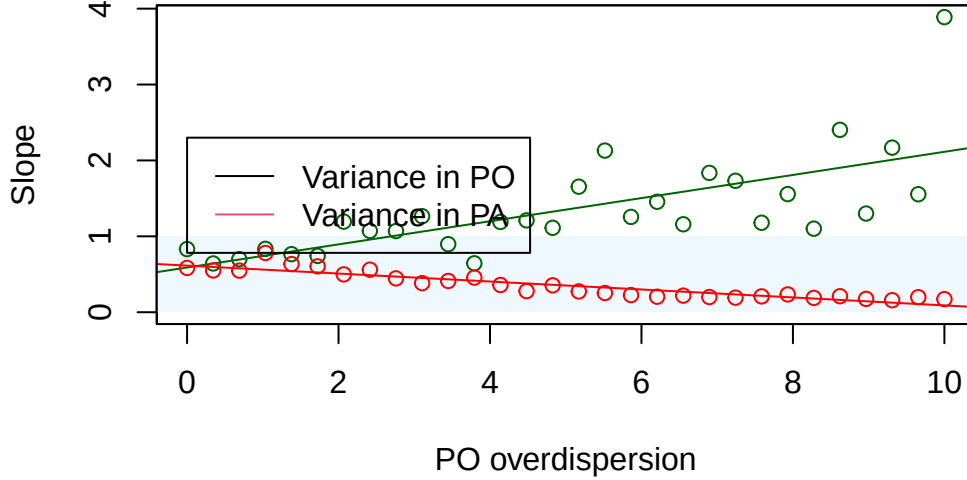


Figure 1: Estimated slopes for calibration of predictions to simulated presence absence data, from Maxent model with linear features fitted to simulated PO data with overdispersion in the linear predictor.

Reality will not be this clean, of course. γ_i and δ_i will be correlated, which will bias the estimation of γ_i , i.e. the niche. And of course, there will be observation uncertainty in both the PA and PO data. In addition, even if the model is correct, there will be estimation uncertainty, which will act like δ_i , and will attenuate the coefficients towards zero (Carroll et al. (1984)).

Overall, if we expect the MaxEnt model is correct, the regression coefficient should be below 1. If it is wrong, the coefficient could be greater or smaller than 1. A negative coefficient would indicate a spectacularly bad model.

Methods

Data

Six data sets from the `disdat` R package were used (Elith et al. (2020)). Each data set contains a presence-only data set, along with randomly chosen background points, and a presence/absence data set. Each of these has data on xx - yy species. As well as the species observations, the data consist of xx - yy environmental covariates, chosen to be relevant for the species groups. Most were continuous, and were standardised to have mean zero and variance 1. For the presence absence data the covariates were standardised using the means and variance of the PO data.

These are:

- AWT:
- CAN:
- NSW:
- NZ:
- SA:
- SWI:

For New Zealand we merged classes 2 (high) and 3 (not documented) of toxicats, and for NSW we merged level 8 of vegsys (pine plantation) with Dry Open Forest. In both cases, these are rare classes that are not in every data set. We only used species with at least 20 PO data points.

Modelling

For each data set we followed the same analysis path. First, a MaxEnt model was fitted to the presence-only data. This used all of the features as a default, but if this model did not converge, a model without hinges was used [CHECK THAT NONE OF THE MODELS WERE REDUCED FURTHER]. For background points, two strategies were used: using the background points given in the data, and using the presences of other species as background points (REF). Only the BEST is presented in the main text: analyses with THE OTHER are reported in the supplementary material. As well as fitting these models, we used five-fold internal cross-validation and calculated AUC and a False Skills Statistic. Cross validation folds were chosen at random or using spatial cross-validation (REF): only the BEST is shown in the main text, with THE OTHER reported in the supplementary information. The MaxEnt model was used to predict propensity on the presence/absence data.

Two (?) models were fitted to the presence-absence data: one model using the propensity from the PO model as a covariate, and a second fitting the MaxEnt model (i.e. the same regularisation and features). This was carried out by changing the weights for the absences from 100 to 1, and keeping the rest of the model the same. Both the cloglog and logit link were tried: only THE BEST is presented here.

The fit of the presence-only model to the presence-absence data was examined in two ways. First, the regression coefficient from the propensity model was examined: it should be close to 1. Second, the correlations of the predictions from the PO and PA data were compared. Even with a correct model these will be below the ideal of 1, because of sampling error in both data sets.

Simulations were used to explore the outcomes when there is no model misfit. For each species a logistic regression model with linear features was fitted to the presence-absence data. This model was then used to simulate presence only and presence-absence data. The presence only data were sampled without replacement from the background points from the data, with a probability proportional to the exponent of the propensity. The same number of points as in the original data were sampled. For the presence absence data the same model was used,

including the same intercept. These data were then analysed with the same models as above [CHECK].

Refactoring

- get data (sim or real)
- fit MaxEnt to PA
 - fit MaxEnt to PO
 - predict & fit to PO
 - * calc. beta & correlation

Results

[NOT SURE WHERE TO PUT THIS] The PA and PO data are supposedly sampled from the same actual distribution, so we would expect there to be strong correlations between internal and external fitted models and validation statistics. Ideally, the only deviation should be due to sampling variation, but of course data from the real world does not repeat our ideals. This means that comparisons with null hypotheses are meaningless: the whole point of this exercise is that we expect a relationship, and the question is how strong it is, and if it is strong enough to be reliable.

Summary of Predictions on PA data

```
GetPars <- function(reglst) {
  tt <- t(list2DF(lapply(reglst, function(l) l$coefficients)))
  colnames(tt) <- c("Intercept", "Slope")
  tt
}
Params <- lapply(AllCoefs.lqpth, GetPars)

par(mfrow=c(2,3), mar=c(3,2,4,1), oma=c(2,2,0,0))
sapply(names(Params), function(nm, lst) {
  plot(lst[[nm]][,"Intercept"], lst[[nm]][,"Slope"], type="n", xlab="", ylab="", main=nm)
  rect(-100, 0, 100, 1, col="mistyrose", border=NA)
  # rect(-100, -100, 0.7, 100, col="mistyrose", border=NA)
  # rect(-100, -100, 0.7, 0.7, col="pink", border=NA)
  # rect(-100, -100, 100, 0, col="red", border=NA)
  # rect(-100, -100, 0.5, 100, col="red", border=NA)
  points(lst[[nm]][,"Intercept"], lst[[nm]][,"Slope"])
  box()
})
```

```
}, lst=Params)
```

```
$AWT  
NULL
```

```
$CAN  
NULL
```

```
$NSW  
NULL
```

```
$NZ  
NULL
```

```
$SA  
NULL
```

```
$SWI  
NULL
```

```
mtext("Intercept", 1, outer=TRUE)  
mtext("Slope", 2, outer=TRUE, line=0.5)
```

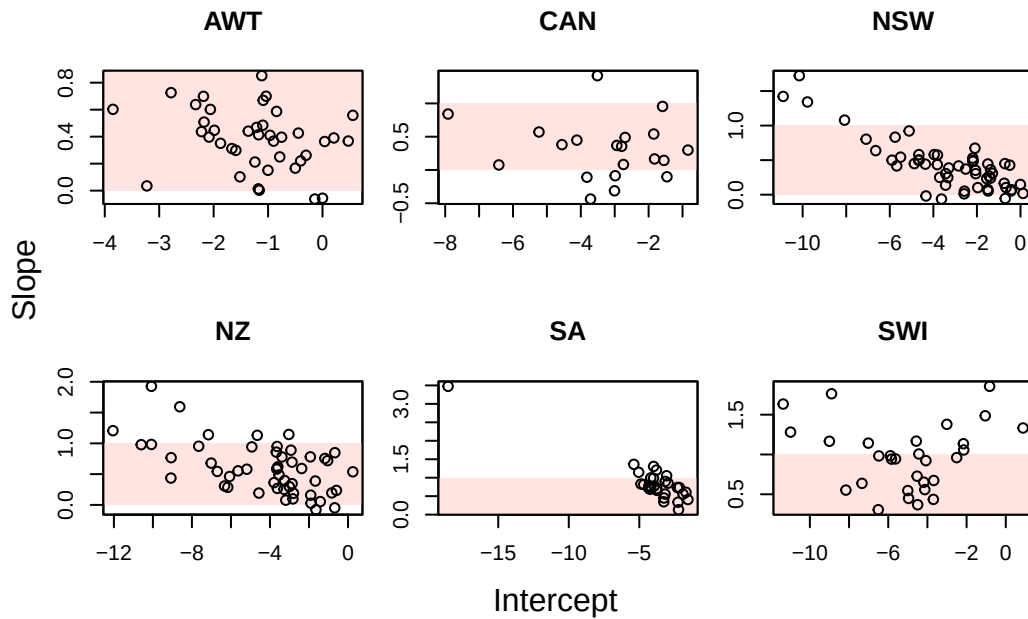


Figure 2: Ung

The estimated slopes and intercepts of the GLM linking the predictions from the MaxEnt model to the PA data are shown in Figure 2. Although most of the slopes (69%) are between 0 and 1, as we might expect, 3% were negative, and 28% had slopes above 1, with 16% not having 1 in the confidence interval. If we view an estimated slope between 0.5 and 1 as acceptable, only 39% of species fit this category.

Comparison with Traditional Indicators

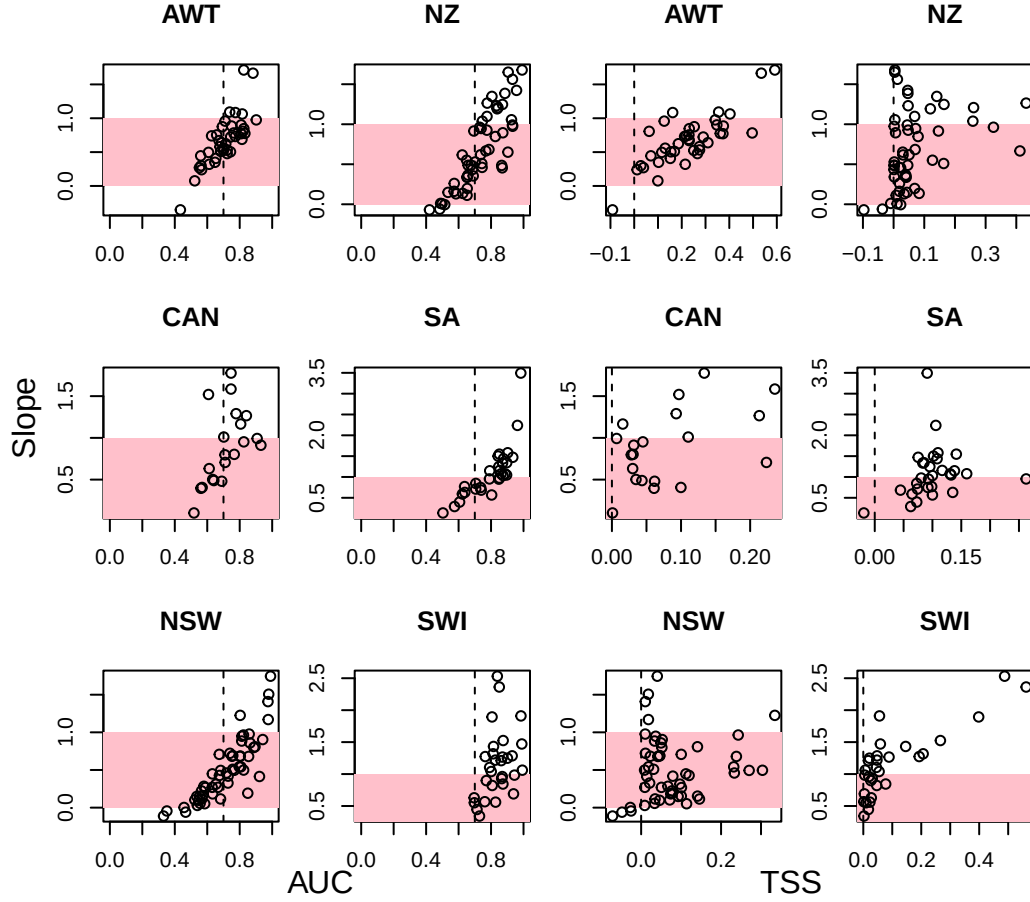


Figure 3: Estimated intercept and slope for calibration of predictions to presence absence data, from Maxent model with all features.

In general there is a positive correlation with AUC (Figure 3), but smaller correlations with TSS. As with AUC, the slope can tell us which species have bad models, i.e. where treating a predicted presence as an absence is more likely to be correct. Overall the data, 35% of the species have an AUC below 0.7, and 4% have an AUC below 0.5.

Internal Cross-validation

The regression coefficients are not well predicted by the AUC from external cross-validation (Figure 5), with the highest correlation being 0.42. In part this may be because AUC cannot distinguish between a small regression coefficient and an ideal one, but where there is still much uncertainty. THE EFFECT OF TSS IS....

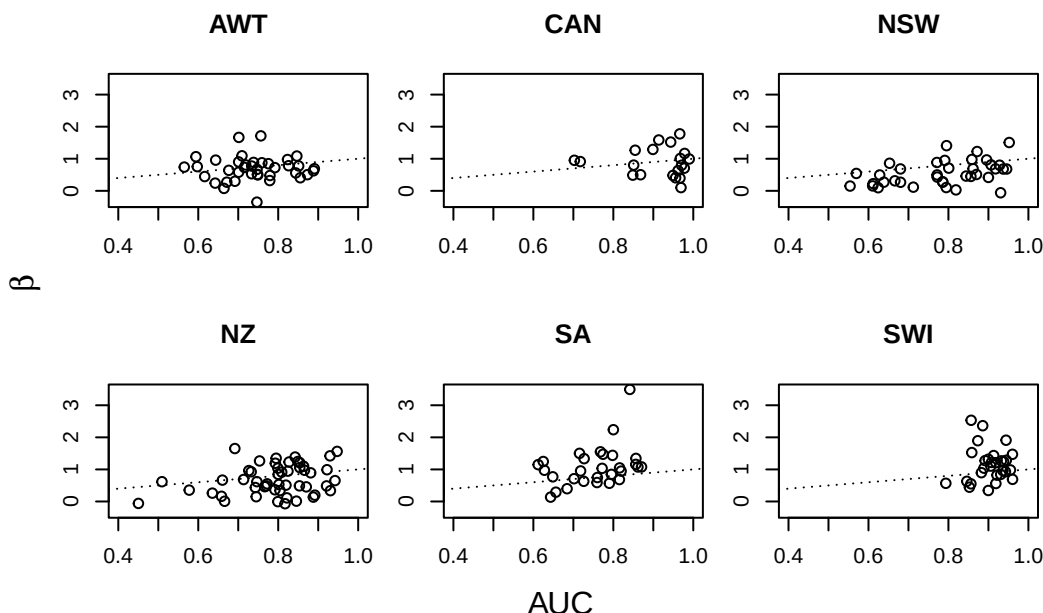


Figure 4: Comparison of regression coefficient and AUC from external cross-validation.

The relationship between internal and external cross-validation statistics is mostly weak or moderate (Figure 5), with only the Swiss data having an almost-strong correlation.

Correlations between models fitted to PA and PO data

Most of the correlations between the models fitted to the PA and PO data were weakly positive, with 1% being negative, and 44% having a correlation below 0.6 (Figure 6). This contrasts with the simulated data where 13% of the simulated correlations were below 0.6. [CHECK THIS!!! SEEMS TO BE WRONG]

Plots of predictions from the MaxEnt model on the validation data model plotted against predictions for a model with MaxEnt features fitted to the presence-absence data.

In the simulated data, two species (nz21 & swi12) have negative correlations between the fitted models. We can see in the plots of the parameter estimates that mostly lie along one or two of the axes (Figure 8), i.e. the regularisation shrinks the estimates to zero, and for these species different coefficients remain in the models fitted to the two data sets. It must be emphasised

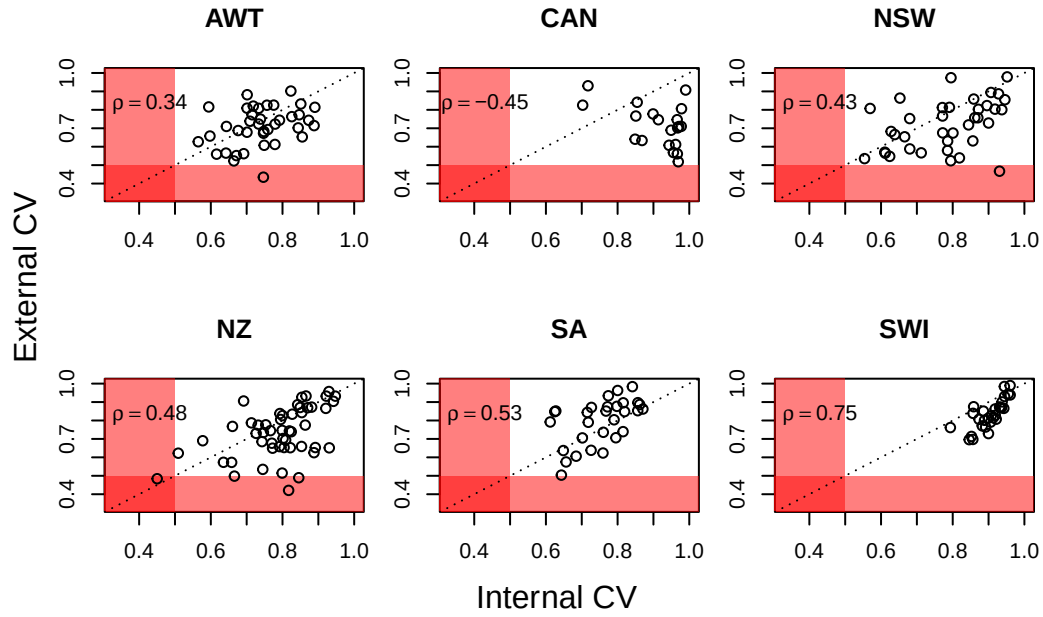


Figure 5: AUC from internal and external validation

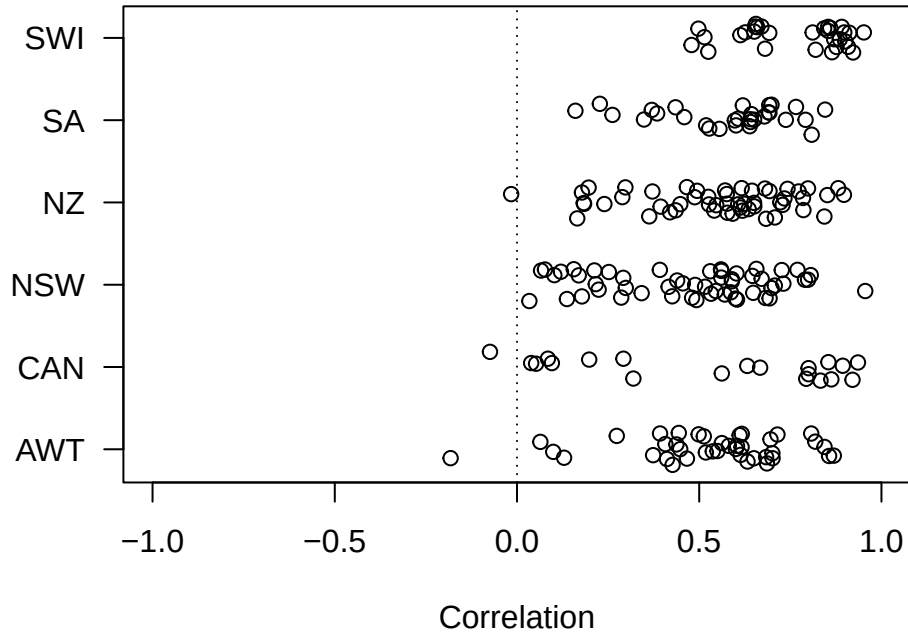


Figure 6: Correlations between predictions of Maxent models fitted to presence-only data and presence-absence data, predictions made on presence-absence locations

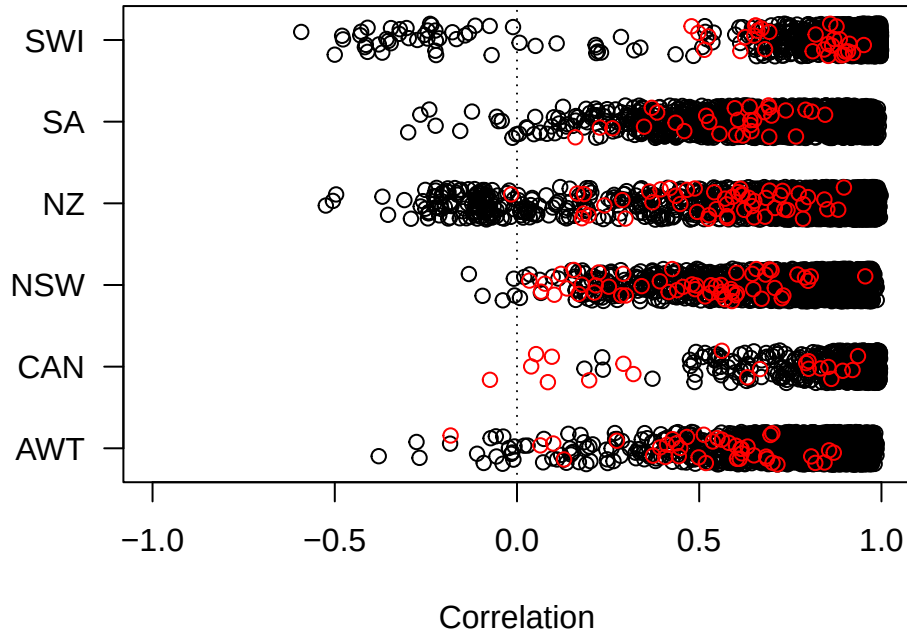


Figure 7: Correlations between predictions of Maxent models fitted to presence-only data and presence-absence data, predictions made on presence-absence locations

that these are extreme data: the behaviour is instructive, as an extreme example of how the models can fail rather than examples of typical behaviour.

Simulated Data

(NOT SURE THIS SHOULD GO HERE)

With the simulated data, 87% of the simulated correlations (and 88% of the mean correlations) were above the correlations in the data (Figure 9). The models where the simulated data performed worse tended to be those with less data (Figure 10). This is what might be expected.

When we look at beta, we see that it varies a lot in the simulated data, even though the data are from the same process. Because the model is correct, but the uncertainties in the PO model are ignored, we would expect the average beta to be below 1, but this is only true for 49% of the data sets. We can see that as the number of presence-only points rises, the variation in beta gets smaller, but the average increases (Figure 11). Thus, with more data the average beta becomes more biased.

```
YLim <- c(min(Betas$meanbeta-Betas$sdbeta, na.rm = TRUE),
          max(Betas$meanbeta+Betas$sdbeta, na.rm = TRUE))
```

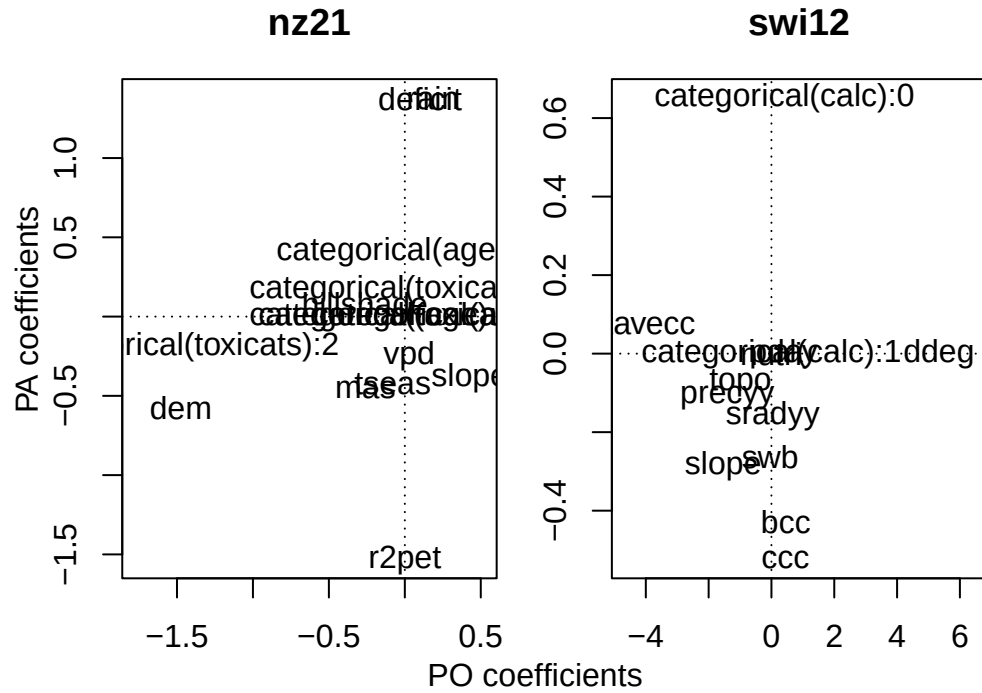


Figure 8: Estimates of parameters from fits of MaxEnt models to presence-only and presence absence data for species nz21 and swi12.

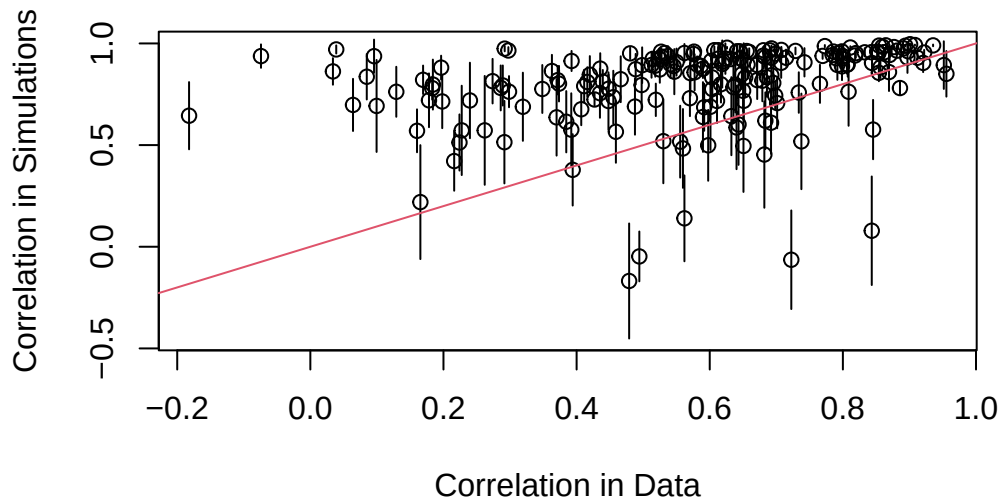


Figure 9: Relationship between correlations between predictions on PA data for MaxEnt models fitted on PO and PA data. Simulated correlations are mean and mean $\pm$ 1 standard deviation.

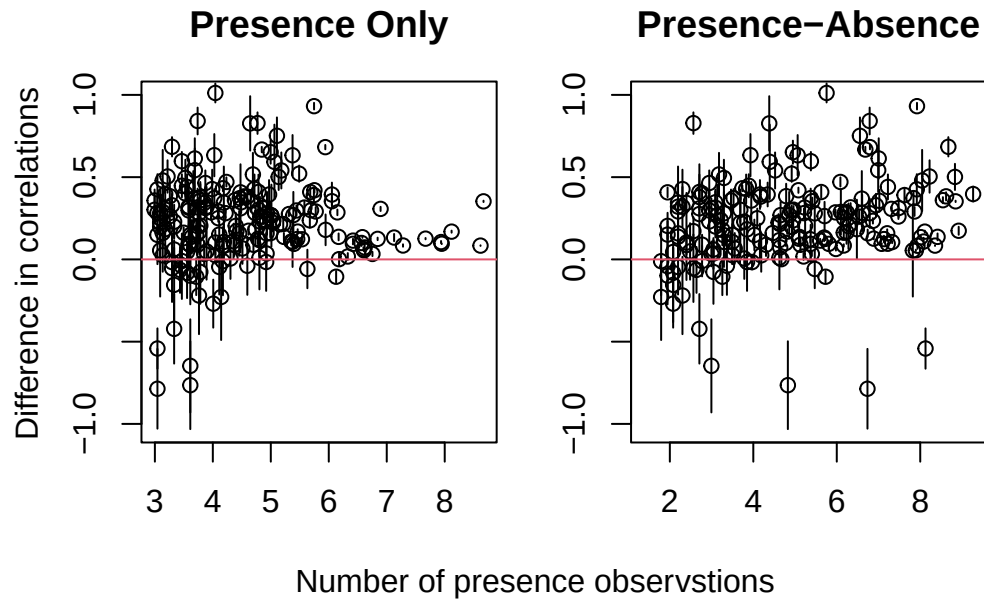


Figure 10: Effect of number of observations of a species on the difference in correlations.

```
par(mfrow=c(1,2), mar=c(3.1,2.5,3,1), oma=c(2,2,0,0))
plot(log(Betas$nPO), Betas$meanbeta, ylim=YLim,
      xlab="", ylab="", main="Presence Only")
segments(log(Betas$nPO), Betas$meanbeta-Betas$sdbeta,
          log(Betas$nPO), Betas$meanbeta+Betas$sdbeta)
abline(h=1, col=2)

plot(log(Betas$nPA), Betas$meanbeta, ylim=YLim,
      xlab="", ylab="", main="Presence Absence")
segments(log(Betas$nPA), Betas$meanbeta-Betas$sdbeta,
          log(Betas$nPA), Betas$meanbeta+Betas$sdbeta)
abline(h=1, col=2)

mtext("Betas", 2, outer=TRUE)
mtext("Number of presence observations", 1, outer=TRUE)
```

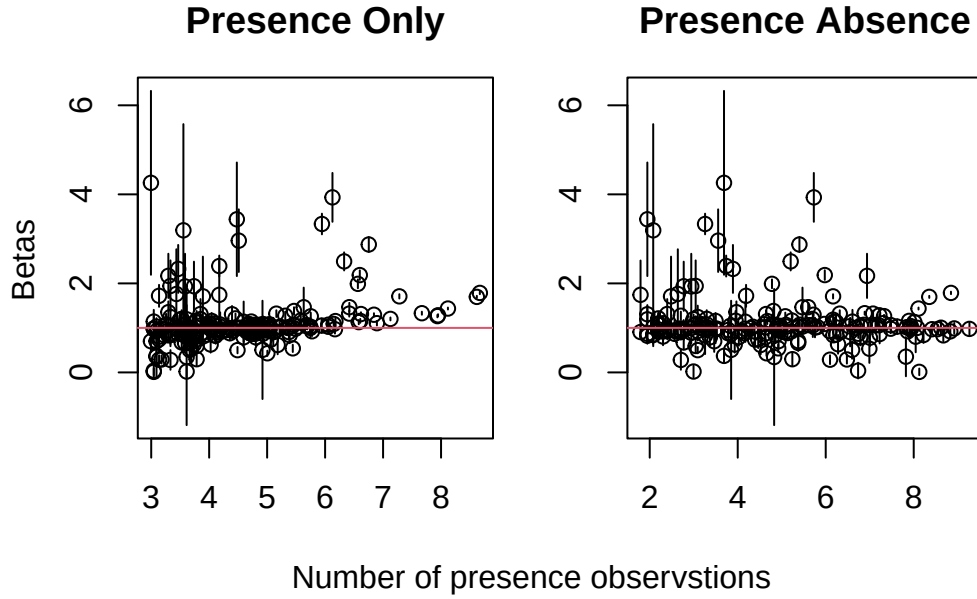


Figure 11: Relationship between betas for MaxEnt models fitted on PO and PA data. Simulated betas are mean and mean $\pm$ 1 standard deviation.

Variation in Prevalences

As would be expected, the estimated intercepts varied between species, i.e. species had differing observed prevalences. This was not correlated between data sets: the largest correlation is 0.32, for the NZ data.

The MaxEnt model needs an intercept, but this is determined by the number of presence and pseudo-absence points, so it is not generally helpful. However, we might hope that the relative values of intercepts across species represent relative measures of prevalence. We can examine this by asking whether the correlations between the intercepts from the PA data are similar to those in the PO data. Unfortunately they are not: the largest correlation is 0.32, for the NZ data.

Discussion

Traditional assessment of SDMs has relied on calculating standard statistics, such as AUC or FSS, through internal or external validation. Here we have shown that it is fruitful to examine the validation more deeply, not just to use validation statistics to compare model, but also to learn more about how badly the models fit, and where problems might arise. In particular, we show that the fitted models often fail, producing predictions that are either counter-intuitive or wrong. The difference between our regression coefficient and 1 can be interpreted as a measure

of model mis-fit, and whilst it might seem strange that it could be above 1, we have seen that this can occur because the PO data has excessive observational noise.

As well as showing the advantages of using a regression approach, this work also underlines the necessity for external cross-validation. The relationships between internal and external cross-validation statistics is weak (see also REFS), and because we can compare the two, we are able to see that one reason for this is the quality of the PO data.

But why should our results be so wrong? We can see this as the confluence of two statistical aphorisms: Box’s famous “All models are wrong, but some are useful” (Box (1979)), and Abelson’s first law of statistics: chance is lumpy (Abelson (1995)). The lumpiness of chance means that the different covariates are correlated, as is common with spatial data (Dormann (2007)). So, because the model is wrong, it is easy for it to select the wrong covariates to explain the data: we can see this for nz21 and swi12 where, even when the true model is one of the candidate models, fits to the two different data sets chose different covariates. The regularisation through MaxEnt’s L1 norm probably accentuates this: without it, the final model would be more of a compromise between the predictors. Selecting the wrong covariate is not a problem for prediction, as long as the correlation with the “true” covariate is stable across all possible data (REF). But, as we see here, the models can be wildly different if this changes.

Although this work suggests that SDMs using MaxEnt are poor, it also helps us see where improvements might be helpful. The regression coefficients above 1 indicate that the fit of the model to the PO data is poor, and thus this is an area where more work would be helpful. This would require understanding more about how the data were collected, starting with whether it is PO or actually PA data. For example, most bird data comes from eBird, whose research grade data consists of complete lists. Thus it can be treated as PA data (or detection/non-detection data might be a better descriptor). Even when the data are PO data, some more investigation will help identify the possible biases, e.g. if the detections are along specific routes (REF). To find this out requires meta-data about the collection and properties of the data. Extracting this metadata may become easier as the Humboldt extension to Darwin Core standards becomes used (REF), as this will help standardise the metadata, making it easier to find out (for example) where the absences are. Statistical models and approaches to deal with the richer class of data exist (e.g. OCCUPANCY MODELS): one useful approach is to consider the sampling issue as a missing data problem (Bowler et al. (2025)).

As a by-product, we have also replicated the findings of Smith and Levine (2025). The intercepts for the PA data were not 0, i.e. most species did not have a prevalence over the region of 0.5, and the prevalences varied, i.e. some species are more widespread than others. Whilst this should not be surprising, it does emphasise that the “arbitrary” (REF) choice of $\tau = 0.5$ is not just arbitrary but, in practice, wrong. Calibration of species’ distributions thus needs to be done with more care. Unfortunately the lack of correlation between the intercepts in the PO and PA data suggests we cannot just use models of PO data for many species and correct all of the relative prevalences up or down.

Uncertainty in the distribution models has largely been ignored. This is in common with most SDMs, and also because standard errors and prediction intervals are not easy to obtain from regularised models, an issue made worse because the statistical theory underlying presence-only data shows that resampling methods such as internal cross-validation or bootstraps are not guaranteed to work (REF AND CHECK). Sampling the likelihood, or being Bayesian and sampling the posterior distribution, is possible but slow, because of the large number of parameters.

The work here agrees with A. Lee-Yaw et al. (2022) that SDMs that are commonly used often behave poorly. But it also points to pathways to improvement. Finding better models for the actual distribution will only bring marginal gains, at best. Better improvements can be made by taking the data collection process more seriously: the analysis here suggests that this is where a large part of the problem is. This extra work will make it more difficult to fit SDMs to large numbers of species, but the gain in producing reliable models will surely be worth it.

References

```
# Want to do this at the end
library(quarto)
quarto_render("ValidatingSDMsSupplInfo.qmd") # all formats
```

processing file: ValidatingSDMsSupplInfo.qmd

```
|
|                                     | 0%
|
|..                                 | 3%
|
|...                               | 6% [Setup]
|
|.....                             | 10%
|
|.....                             | 13% [fig-Intercepts]
|
|.....                             | 16%
|
|.....                             | 19% [fig-GoF]
|
|.....                             | 23%
|
```


.....	26% [IntCVTSS]
.....	29%
.....	32% [Diffs]
.....	35%
.....	39% [fig-PlotCorrDiff]
.....	42%
.....	45% [fig-PlotCorrDiffN0bs]
.....	48%
.....	52% [fig-Corrslqpt.sp]
.....	55%
.....	58% [MaxEntPAPredsAWT]
.....	61%
.....	65% [MaxEntPAPredsCAN]
.....	68%
.....	71% [MaxEntPAPredsNSW]
.....	74%
.....	77% [MaxEntPAPredsNZ]
.....	81%
.....	84% [MaxEntPAPredsSA]
.....	87%
.....	90% [MaxEntPAPredsSWI]
.....	94%

```
|
|..... | 97% [PLOTcors]
|
|..... | 100%
```

output file: ValidatingSDMsSupplInfo.knit.md

pandoc

```
to: html
output-file: ValidatingSDMsSupplInfo.html
standalone: true
section-divs: true
html-math-method: mathjax
wrap: none
default-image-extension: png
variables: {}
```

metadata

```
document-css: false
link-citations: true
date-format: long
lang: en
title: ValidatingSDMSSupplInfo
author: Bob O'Hara
```

Output created: ValidatingSDMsSupplInfo.html

- A. Lee-Yaw, Julie, Jenny L. McCune, Samuel Pironon, and Seema N. Sheth. 2022. "Species Distribution Models Rarely Predict the Biology of Real Populations." *Ecography* 2022 (6): e05877. <https://doi.org/https://doi.org/10.1111/ecog.05877>.
- Aarts, Geert, John Fieberg, and Jason Matthiopoulos. 2012. "Comparative Interpretation of Count, Presence–Absence and Point Methods for Species Distribution Models." *Methods in Ecology and Evolution* 3 (1): 177–87. <https://doi.org/https://doi.org/10.1111/j.2041-210X.2011.00141.x>.
- Abelson, R. P. 1995. *Statistics as Principled Argument*. L. Erlbaum Associates. <https://books.google.no/books?id=TgmbosIA7N0C>.
- Bowler, Diana E., Robin J. Boyd, Corey T. Callaghan, Robert A. Robinson, Nick J. B. Isaac, and Michael J. O. Pocock. 2025. "Treating Gaps and Biases in Biodiversity Data as a Missing Data Problem." *Biological Reviews* 100 (1): 50–67. <https://doi.org/https://doi.org/10.1111/brv.13127>.
- Box, G. E. P. 1979. "Robustness in the Strategy of Scientific Model Building." In. <https://api.semanticscholar.org/CorpusID:121999337>.
- Carroll, Raymond J., Clifford H. Spiegelman, K. K. Gordon Lan, Kent T. Bailey, and Robert D. Abbott. 1984. "On Errors-in-Variables for Binary Regression Models." *Biometrika* 71

- (1): 19–25. <http://www.jstor.org/stable/2336392>.
- Dormann, Carsten F. 2007. “Effects of Incorporating Spatial Autocorrelation into the Analysis of Species Distribution Data.” *Global Ecology and Biogeography* 16 (2): 129–38. <https://doi.org/https://doi.org/10.1111/j.1466-8238.2006.00279.x>.
- Elith, Jane, Steven J. Phillips, Trevor Hastie, Miroslav Dudík, Yung En Chee, and Colin J. Yates. 2011. “A Statistical Explanation of MaxEnt for Ecologists.” *Diversity and Distributions* 17 (1): 43–57. <https://doi.org/https://doi.org/10.1111/j.1472-4642.2010.00725.x>.
- Elith, J., Graham, C.H., Valavi, R., Abegg, et al. 2020. “Presence-Only and Presence-Absence Data for Comparing Species Distribution Modeling Methods.” *Biodiversity Informatics* 15: 69–80. <https://doi.org/10.17161/bi.v15i2.13384>.
- Fithian, William, and Trevor Hastie. 2013. “Finite-sample equivalence in statistical models for presence-only data.” *The Annals of Applied Statistics* 7 (4): 1917–39. <https://doi.org/10.1214/13-AOAS667>.
- Phillips, Steven, and Miroslav Dudík. 2008. “Generative and Discriminative Learning with Unknown Labeling Bias.” In *Advances in Neural Information Processing Systems*, edited by D. Koller, D. Schuurmans, Y. Bengio, and L. Bottou. Vol. 21. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2008/file/9cf81d8026a9018052c429cc4e56739b-Paper.pdf.
- Renner, Ian W., and David I. Warton. 2013. “Equivalence of MAXENT and Poisson Point Process Models for Species Distribution Modeling in Ecology.” *Biometrics* 69 (1): 274–81. <https://doi.org/https://doi.org/10.1111/j.1541-0420.2012.01824.x>.
- Smith, Jeffrey R., and Jonathan M. Levine. 2025. “Linking Relative Suitability to Probability of Occurrence in Presence-Only Species Distribution Models: Implications for Global Change Projections.” *Methods in Ecology and Evolution* 16 (4): 854–65. <https://doi.org/https://doi.org/10.1111/2041-210X.70003>.
- Valavi, Roozbeh, Gurutzeta Guillera-Arroita, José J. Lahoz-Monfort, and Jane Elith. 2022. “Predictive Performance of Presence-Only Species Distribution Models: A Benchmark Study with Reproducible Code.” *Ecological Monographs* 92 (1): e01486. <https://doi.org/https://doi.org/10.1002/ecm.1486>.
- Warton, David I., and Leah C. Shepherd. 2010. “Poisson point process models solve the ‘pseudo-absence problem’ for presence-only data in ecology.” *The Annals of Applied Statistics* 4 (3): 1383–1402. <https://doi.org/10.1214/10-AOAS331>.
- Yackulic, Charles B., Richard Chandler, Elise F. Zipkin, J. Andrew Royle, James D. Nichols, Evan H. Campbell Grant, and Sophie Veran. 2013. “Presence-Only Modelling Using MAXENT: When Can We Trust the Inferences?” *Methods in Ecology and Evolution* 4 (3): 236–43. <https://doi.org/https://doi.org/10.1111/2041-210x.12004>.